

UML Exercise: Design a Library Management System

Problem Statement:

You are required to design a **Library Management System** using UML diagrams. The system should manage books, library members, and the borrowing process. The system should have the following capabilities:

1. **Books:**
 - Each book should have a title, author, ISBN, and publication date.
 - Books can be available in different genres (e.g., Fiction, Non-Fiction, Science, etc.).
 - The library keeps track of the number of copies of each book.
2. **Members:**
 - Library members should have a name, membership ID, contact details, and membership expiration date.
 - Members can borrow books from the library.
 - Each member can borrow up to 5 books at a time.
 - The system should track which books are currently borrowed by each member.
3. **Borrowing Process:**
 - A member can borrow a book if it is available in the library.
 - The borrowing record should include the borrow date and the due date for return.
 - Members should be able to return books, and the system should update the availability of the book.
4. **Fines:**
 - If a member returns a book after the due date, they should be fined.
 - The fine should be calculated based on the number of days overdue.

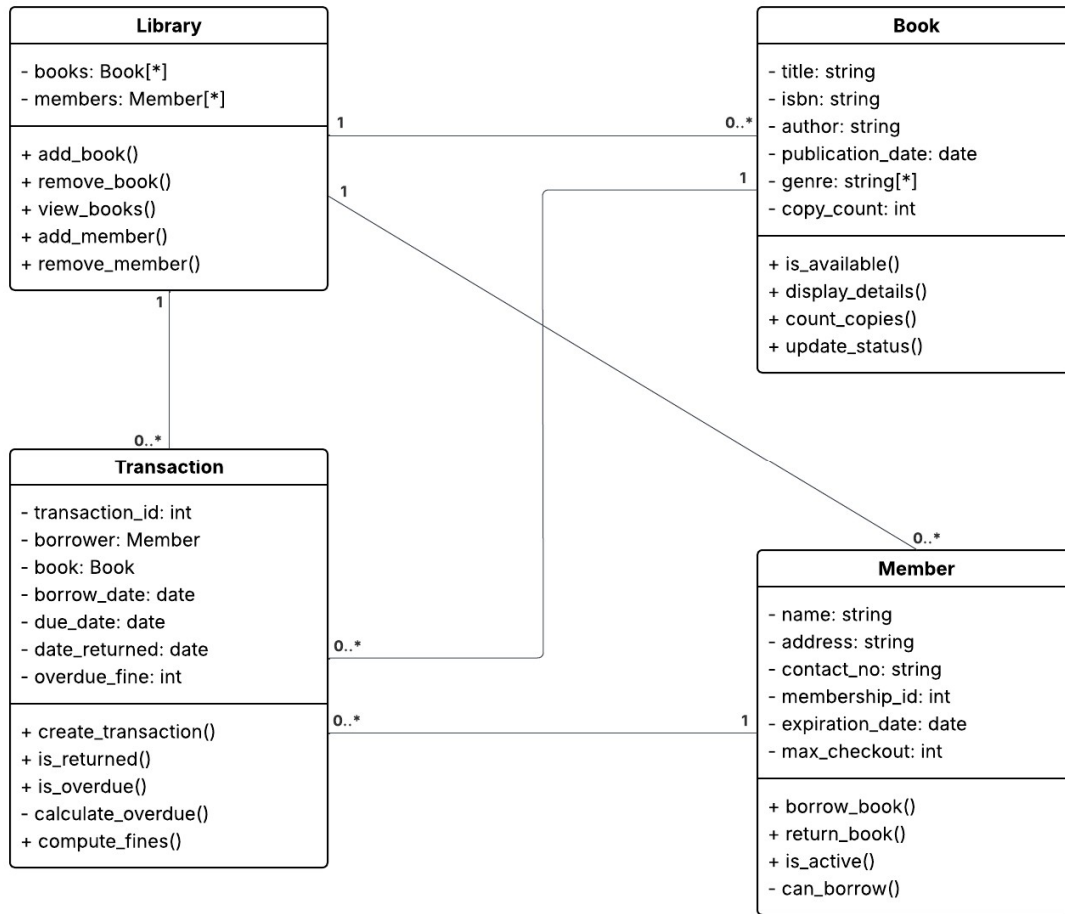
Tasks:

1. **Class Diagram:**
 - Create a class diagram to represent the system, showing the classes, their attributes, methods, and the relationships between them.
2. **Use Case Diagram:**
 - Draw a use case diagram to identify the main actors (e.g., Librarian, Member) and the key use cases (e.g., Borrow Book, Return Book, Add New Book).
3. **Sequence Diagram:**
 - Create a sequence diagram to illustrate the process of a member borrowing a book, including interactions between the member, the book, and the library system.
4. **Activity Diagram:**
 - Design an activity diagram to show the workflow of returning a book, including the steps of checking for overdue fines and updating the book's availability.

Deliverables:

- Create your UML diagrams (Class Diagram, Use Case Diagram, Sequence Diagram, and Activity Diagram) with proper annotations.
- Provide a brief explanation of each diagram, describing the design decisions and how the system fulfills the requirements.

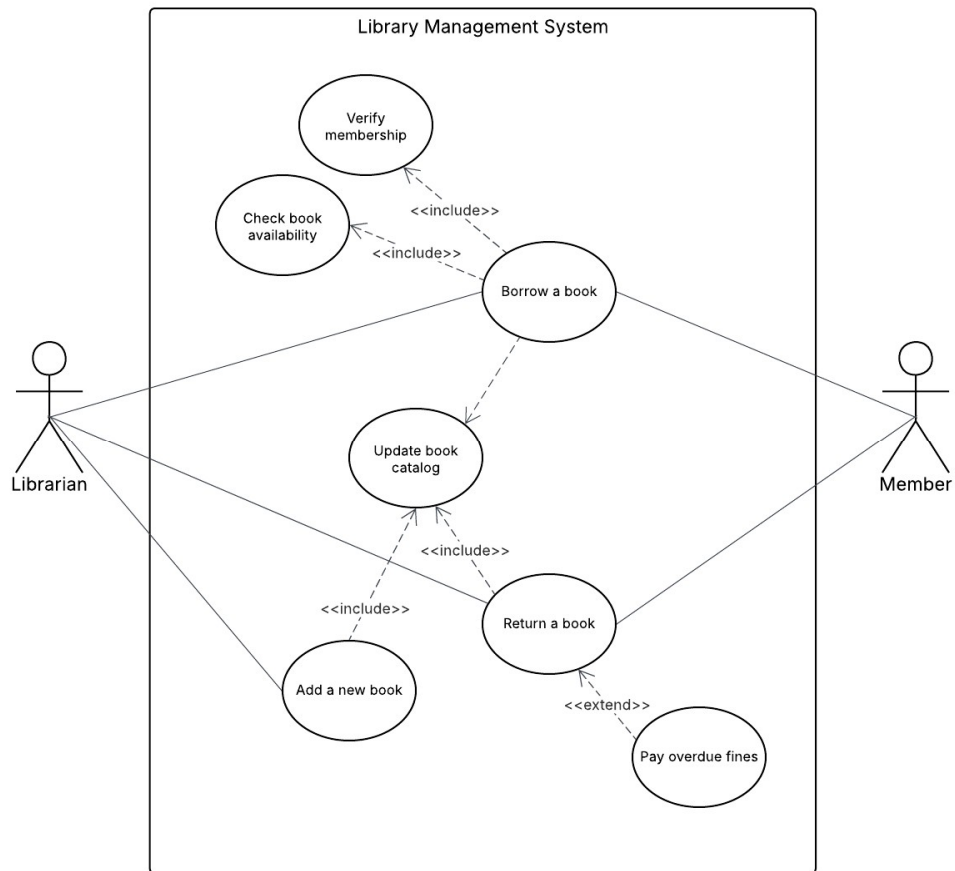
1. Class Diagram



Relationships:

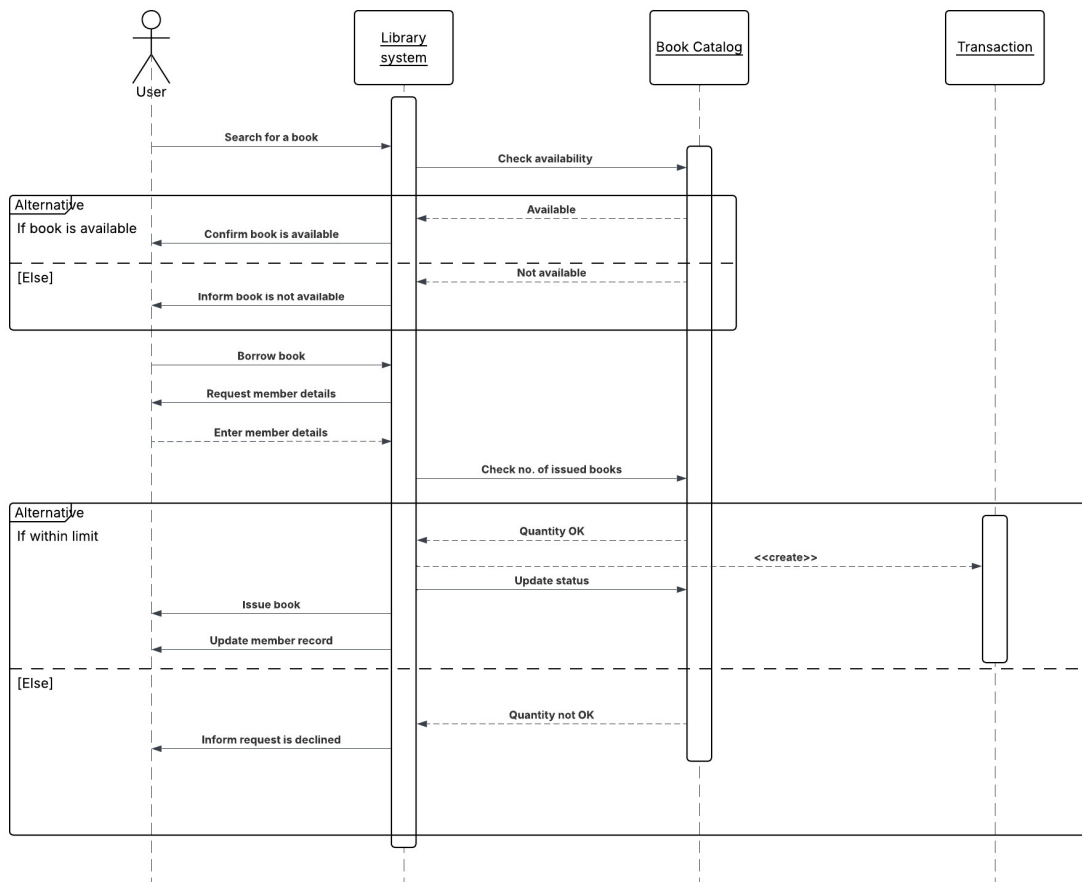
- A library has books and members
- A member belongs to a library where the member can borrow books from.
- A transaction is a record of the member's checked out books from the library.

2. Use Case Diagram



In this use case diagram, the primary actor is the Librarian, and the secondary actor is the Member. The librarian can process the borrow book request or return book request of a member, and also add new books to the library. The member can borrow or return books.

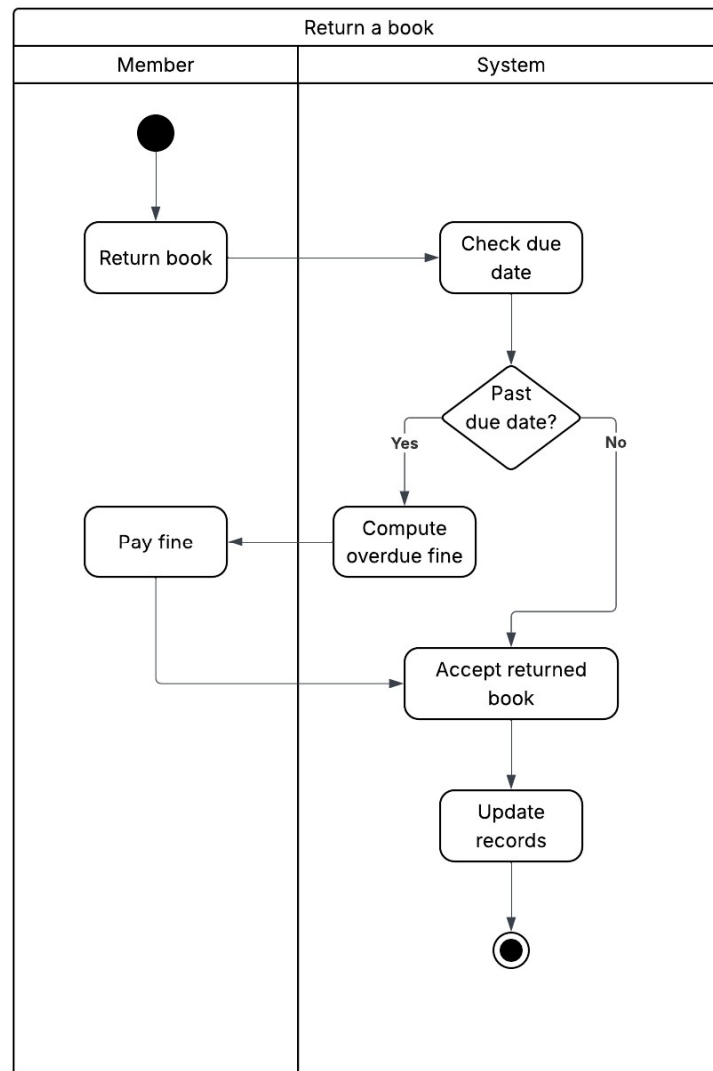
3.. Sequence Diagram



Process flow when borrowing a book:

1. The user (librarian or member) checks if the book planned to borrow is available in the library catalog.
2. The member should provide membership details for verification in the system.
3. The system checks if the member has reached the maximum limit of 5 checked out books at a time.
 - a. If no, then the book can be issued. The system will also create and record the details of the transaction.
 - b. If yes, the book cannot be issued.

4. Activity Diagram



The process for returning a book requires checking if the book has been returned on or before the due date. If the book has been returned late, a fine based on the number of days late must be paid to complete the process.